TSMC

17 April 2026

Leading edge supply tightness through 2027 should drive continued EPS upgrades; Raise PT to NT$2500

We are raising our FY26/27E EPS ests by 4% each and raise our Dec-26 PT to NT $2500 to reflect supply tightness in leading edge process nodes well into 2027 or early 2028. Accelerating AI demand (Accelerators, CPU and networking), continued growth in N3, N2 and advanced back-end revenues and long lead time for capacity build are likely to create a favorable set up for TSMC stock, looking quite favorable in the next few quarters, with likely upside to its long-term capex targets and 2026 revenue guidance, and GMs staying high due to high utilization and ASP uptick. Given industry-wide shortage, noise about competition may remain persistent, but achieving technology and manufacturing milestones and winning customer acceptance would be the key challenges for the chasing pack, given TSM's technology and execution leadership and a 4-5-year window for design and manufacturing ramp. In the very short term, the stock may take a breather given the fast run up ahead of earnings and expectations of a bigger revenue/capex guidance hike, but we expect EPS upgrades to come throughout 2026.

- **Capacity tightness to last into 2027, leading edge capacity (N5 and below) growth at 25% CAGR in 25-28E:** Given the strong AI accelerator, CPU and networking demand and continued strength from Apple iPhone and Mac, we expect the leading edge capacity (N5 and below) tightness to last well into 2027 or early 2028. We expect TSMC to accelerate its capacity growth (JPMe 25% CAGR for N5 and below capacity from 2025-28E), with much of the capacity addition coming in the 2027-28 timeframe. As a result, we expect capex to rise meaningfully in the next two years. Given tight capacity in leading edge, we expect to see another round of like-for-like price increase discussions kickstarting in 2Q26 (for 2027 pricing), which could be a further catalyst for the stock.
- **A slew of guidance raises are coming, given obvious AI demand upside and faster supply ramp:** One of the pushbacks from investors in the near-term is the lack of meaningful guidance upside during the 1Q26 earnings call. We believe some patience is required, since TSMC is likely to update its 2026E revenue growth expectations (likely 35+% USD growth) and AI semis revenue demand growth (likely close to 60% after incorporating categories such as networking, new kinds of accelerators such as LPUs and CPUs going into AI workloads – all categories growing much faster than core AI accelerators). In addition, we also expect TSMC to outline a more robust 3-year capex plan (likely reaching ~$190-200bn capex in 2026-28E) to support the extraordinary demand for AI compute from its customers.
- **Are we at the Gross Margin peak? Too early to draw conclusions:** TSMC has seen strong GM pickup in the last 2 years from 56% levels in 2024 to 66% levels in 2026 and given the large gap between the long-term GM guidance (56% and higher) and current Gross Margins, several investors are concerned whether GMs have peaked. In the near-term, dilution from N2 ramp in 2H26 and increased output from AZ Phase 1 will weigh on margins, but N3

Price Performance

	YTD	1m	3m	12m
Abs	34.5%	13.0%	19.8%	143.9%
Rel	6.3%	1.6%	1.6%	53.1%

Company Data

Shares O/S (mn)	25,930
52-week range (NT$)	2100.0-816.0
Market cap ($ mn)	1,708,641
Exchange rate	31.64
Free float (%)	92.1%
3M ADV (mn)	42.92
3M ADV ($ mn)	2,523.4
Volatility (90 Day)	31
Index	TAIEX
BBG ANR (Buy | Hold | Sell)	40|1|0

Key Metrics (FYE Dec)

NT$ in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	3,809,054	5,320,358	7,023,133	8,190,610
Adj. EBIT	1,936,092	3,029,063	3,858,859	4,554,388
Adj. EBITDA	2,624,188	3,834,219	4,895,538	5,780,913
Adj. net income	1,717,883	2,575,416	3,244,463	3,817,544
Adj. EPS	66.24	99.31	125.11	147.21
BBG EPS	64.73	90.00	111.28	133.39
Cashflow from operations	2,437,716	3,303,735	4,108,887	4,916,998
FCFF	1,292,759	1,834,220	2,048,387	2,856,498
Margins and Growth				
Revenue Growth Y/Y (%)	31.6%	39.7%	32.0%	16.6%
EBIT margin	50.8%	56.9%	54.9%	55.6%
EBIT Growth Y/Y (%)	46.4%	56.5%	27.4%	18.0%
EBITDA margin	68.9%	72.1%	69.7%	70.6%
EBITDA Growth Y/Y (%)	32.2%	46.1%	27.7%	18.1%
Net margin	45.1%	48.4%	46.2%	46.6%
Adj. EPS growth	46.4%	49.9%	26.0%	17.7%
Ratios				
Adj. tax rate	16.0%	17.7%	17.7%	17.7%
Interest cover	NM	NM	NM	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	NM	NM	NM
ROE	35.4%	39.6%	37.0%	33.8%
Valuation				
FCFF yield	2.4%	3.4%	3.8%	5.3%
Dividend yield	1.1%	1.2%	1.6%	2.0%
EV/Revenue	13.7	9.5	7.1	5.8
EV/EBITDA	19.8	13.2	10.1	8.3
Adj. P/E	31.5	21.0	16.7	14.2

Summary Investment Thesis and Valuation

Investment Thesis

We expect TSMC's structural growth drivers to remain very strong as leading-edge supply (N4, N3, and N2) stays tight well into 2027 or early 2028, given accelerating AI demand, continued growth in N3, N2 and advanced back-end revenues, and long lead time for capacity build despite accelerated capex. We believe the set-up for TSMC stock looks quite favorable in the next few quarters, with potential upside to its long-term capex targets, 2026 revenue guidance, and GMs beating due to high UTR and ASP uptick. Given industry-wide shortage, we expect competitive noise to persist, but achieving technology and manufacturing milestones and winning customer acceptance would be the key challenge for the chasing pack, given TSMC tech and execution leadership and 3-5 year window for design and manufacturing ramp.

Valuation

Our Dec-26 PT of NT$2,500 is based on ~20x 12-month forward P/E and reflects better GMs and slightly faster growth. Our target multiple is higher than TSMC's five-year average historical multiple.

Performance Drivers

Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.81	0.69
Region: Taiwan	0.90	0.95
Macro:		
HSI Volatility Index	0.12	0.31
JPM EM Currency(EMCI) Fixing	-0.30	-0.27
JP Morgan GBI-EM Global Div	-0.22	-0.23
Quant Styles:		
Value	-0.45	-0.27
DivYld	-0.45	-0.25
Growth	0.35	0.17

Source: J.P. Morgan Global Markets Strategy for Performance Drivers; company data, Bloomberg Finance L.P. and J.P. Morgan estimates for all other tables. Note: Price history may not be complete or exact.

profitability is improving quite fast, due to better product mix (higher HPC and AI accelerator mix) and better yields, while overall utilization and amount of hot-run/super hot-run wafers in N3 are likely to remain high. As a result, we believe that GMs are likely to hover around the mid-high 60% levels, with chances of creating further upside, depending on the extent of price increases in 2027 and a continued tight supply environment. Given depreciation is likely to grow slower than revenues in the next few years, we believe that GMs are likely to remain at elevated levels.

- **Competitive noise likely to stay, but "No Shortcuts" to arriving at production silicon:** TSMC management was quite forceful in its 1Q26 conference call regarding the strong technology, IP ecosystem and manufacturing leadership that TSMC currently possesses at leading edge. A typical leading edge AI accelerator design takes 2-3 years of co-working with the Foundry partner and will then need 1-2 years to eventually enter high volume production – marking a 4-5 year engagement cycle before silicon production. Shortening this process may be possible to some extent with very intense collaboration of a few number of projects, but cost, yield maturity and smooth high volume ramp are all different areas which would take time to master. As such, unless TSMC slows down its technology curve and execution speed (we see the opposite), we do not expect any meaningful challenge to its market leadership in the next 3-5 years. However, the competitive noise is likely to persist, since customers are likely to engage all available options in a period of extreme capacity shortage.
- **Blended ASP growth should bounce back to ~20% levels:** In the last couple of quarters, blended ASP growth has slowed down (to 8% /11% YoY in 4Q25/1Q26). We believe this is due to stronger demand for some of the mature process nodes (16nm, 65nm for CoWoS processing etc.), which is driving stronger wafer shipment growth. As we move through the year, we believe that blended ASP growth is likely to recover to 27% YoY levels in 4Q26, given higher mix of N3 and N2 wafer revenues. Long-term, we believe that TSMC is likely to maintain 15-20% blended ASP growth, while wafer shipments are likely to grow in the 5-10% range in steady state. Better pricing helps TSMC earn a higher return of the increased capex, while also smoothing down big swings in revenues (since wafer shipments are usually more volatile during a downturn).

Capacity tightness to last into 2027, leading edge (N5 and below) capacity growth at 25% CAGR in 25-28E

We expect leading-edge capacity to remain tight well into 2027 or early 2028 (JPMe N5 and below process node UTR to maintain 100% or above in coming years), driven by:

1. Strong AI Accelerator Demand - We expect shipments of key AI accelerator chips —including NVDA GPU, AMD GPU, Google TPU, AWS Trainium, Meta MTIA, and MSFT MAIA—to grow by 50%+ YoY in 2026 and 35%+ YoY in 2027.
2. Increasing demand for AI interconnect (networking, DSP, retimers, etc.) and Server CPUs, including Google, Amazon, and MSFT in-house chips, as well as AMD/ NVDA CPUs
3. Continued strength from Apple iPhone and Mac processor.

Given the ongoing shortage of leading-edge wafers, we anticipate TSMC will accelerate its capacity expansion, especially for N3 and N2 nodes (JPMe 25% CAGR for N5 and below capacity from 2025-28E), with much of the capacity addition coming 2027-28.

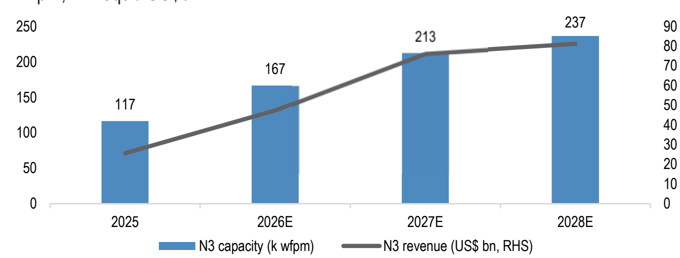
For N3, TSMC plans three new fabs (Tainan Fab 18 P9, AZ Phase 2, and a second Japan fab), ramping through 1H27 into 2028, complemented by cross-fab operations (e.g., leveraging N7 for N3 back-end metal layers and 28nm for N5 back-end metal layers). We expect N3 capacity to grow 28%/11% to reach 213k/237k wfpm by YE26/YE27.

For N2, a robust pipeline—all Apple iPhone models (A20/A20 Pro), AMD Venice and MI450, and flagship SoCs from MediaTek and Qualcomm in 2026, followed by major AI accelerators including Google TPU, NVIDIA Feynman, and Trainium 4 from late 2027–28—should drive a fast ramp. We model N2 capacity reaching ~150k wfpm by end-2028 (vs. ~60k by end-2026), led by Kaohsiung P1–5 and AZ Phase 2.

Due to tight leading-edge capacity, we anticipate another round of price increase discussions in 2Q26 for 2027 pricing (following a 6–10% price hike for leading-edge nodes in early 2026). Our early view is that TSMC could implement an additional 4–5% like-for-like price hike for leading-edge nodes in 2027, which could serve as a further catalyst for the stock.

Figure 1: TSMC N3 capacity build and revenues

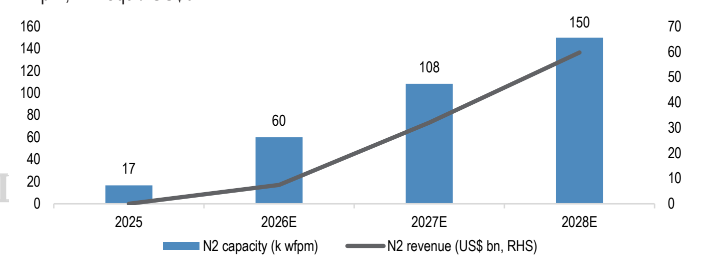
k wfpm, 12" equ / US$bn



Source: J.P. Morgan estimates, Company data.

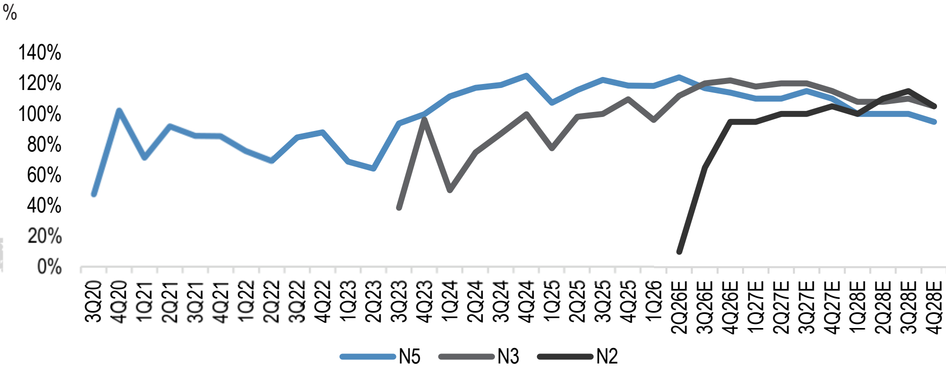
Figure 2: TSMC N2 capacity build and revenues

k wfpm, 12" equ / US$bn



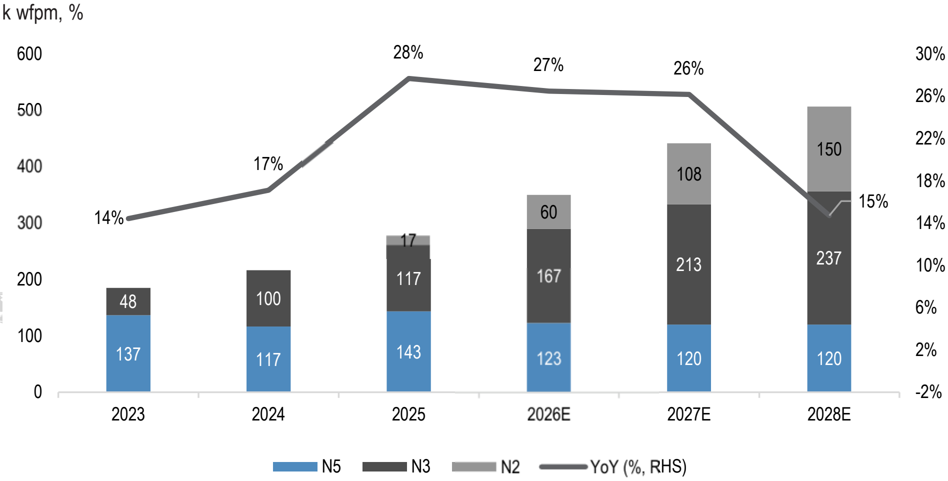
Source: J.P. Morgan estimates, Company data.

Figure 3: Leading edge node UTR since 2020 (N5 launch)



Source: J.P. Morgan estimates, Company data.

Figure 4: TSMC Leading edge (N5 and below) capacity and YoY



Source: J.P. Morgan estimates, Company data.

A slew of guidance raises are coming, given obvious AI demand upside and faster supply ramp

A key near-term pushback from investors is the lack of meaningful guidance upside during the 1Q26 earnings call.

We believe some patience is required, but expect an upward revision of guidance at the upcoming Jul earnings call. We expect TSMC to update its 2026E revenue growth expectations to likely 35+% USD growth (after revising it from close to 30% YoY to above 30% YoY on the 1Q earnings call), and Datacenter AI revenue growth to close to 60% (vs. current guidance of mid to high 50%) after incorporating categories such as networking, new kinds of accelerators (such as LPUs), and CPUs going into AI workloads – all categories growing much faster than core AI accelerators.

Figure 5: TSMC Current guidance vs potential revisions in Jul

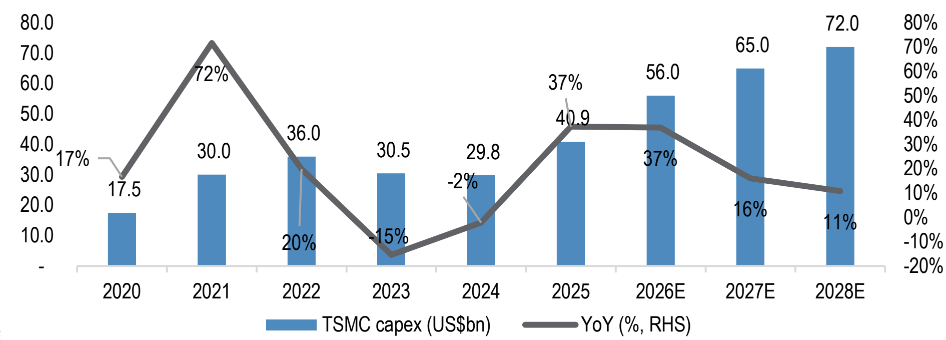
	April Earnings call guidance	July Earnings call (likely guidance)
2026 Rev growth (USD)	>30% YoY	35%+ YoY
LT AI growth CAGR	mid to high 50%	close to 60%
LT Rev growth CAGR	25%	25%
Structural GM	56% and higher	56% and higher

Source: Company data, J.P. Morgan estimates.

In addition, we also expect TSMC to outline a more robust 3-year capex plan of $190-200bn capex for 2026-28E (JPMe $193bn), nearly twice the $101 billion spent in 2023–2025, to support the extraordinary demand for AI compute demand from its customers.

Figure 6: TSMC capex trend

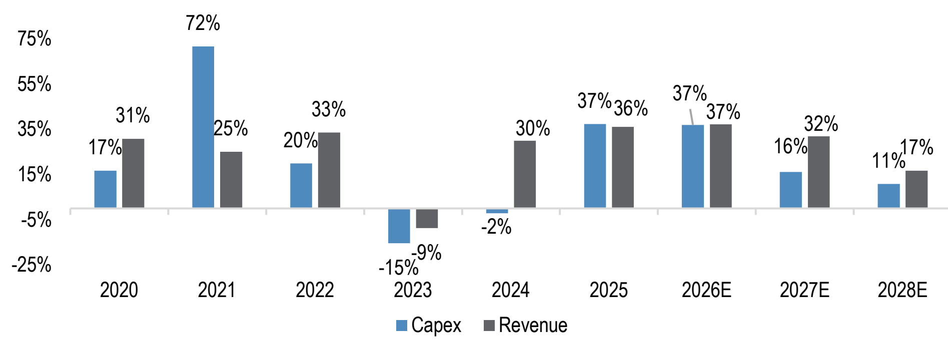
US$bn / %



Source: Company data, J.P. Morgan estimates.

Figure 7: TSMC capex vs. revenue YoY trend

US$bn



Source: J.P. Morgan estimates, Company data.

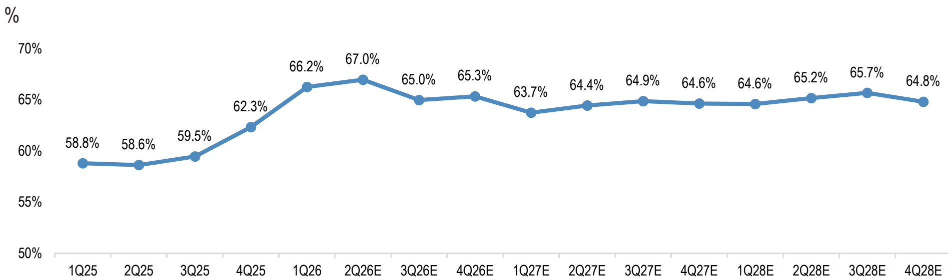
Are we at the Gross Margin peak? Too early to draw conclusions

TSMC has delivered a strong GM uplift over the past two years, moving from the 53–55% range to 65%+, but the wide gap between its long-term GM framework (56% and higher) and current margins has led investors to question whether GMs have peaked.

In the near-term, dilution from N2 ramp in 2H26 and increased output from AZ Phase 1 will weigh on margins. However, we see offsetting positives from fast improving N3 profitability, supported by 1) a richer mix (higher HPC and AI accelerator exposure), 2) better yields, and 3) sustained high utilization (100+% UTR in late 2025-28) with elevated hot/super-hot run intensity.

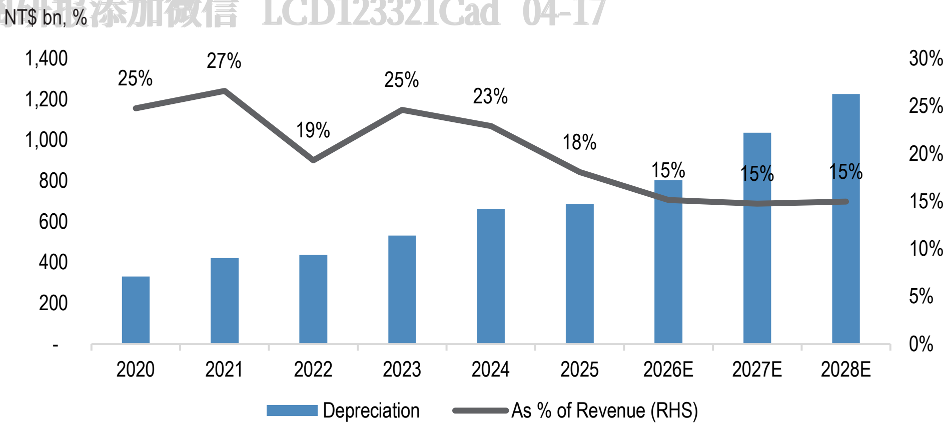
As a result, we expect GMs to hover in the mid-to-high-60% range (JPMe GM 64.4%/65.1% in 2027/28), with potential upside depending on the magnitude of further price increases in 2027 (we believe another 4-5% hike across N5 and below is possible) and a continued tight supply environment. With depreciation expected to grow slower than revenue over the next few years, we believe TSMC can sustain elevated GMs even as new nodes ramp.

Figure 8: GM trend by quarters



Source: J.P. Morgan estimates, Company data.

Figure 9: TSMC Depreciation and as % of revenue



Source: Company data, J.P. Morgan estimates.

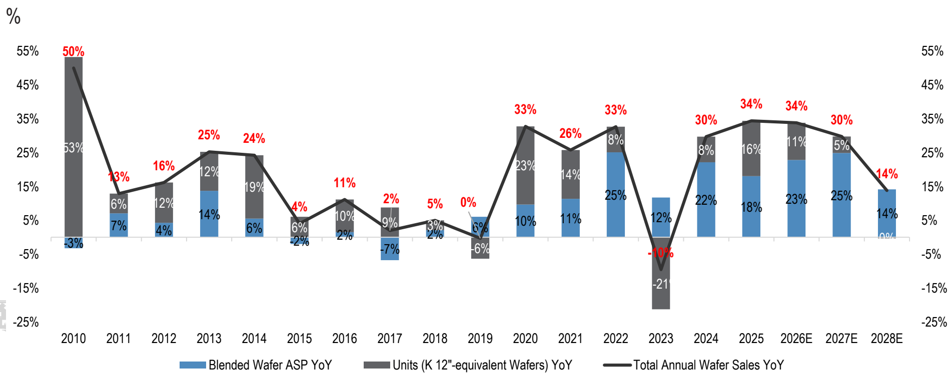
Blended ASP growth, is it slowing? We expect a bounce back to 20% for FY26/27

Blended ASP growth has decelerated to 8% YoY in 4Q25 and 11% in 1Q26, down from ~20% over the prior two years. We attribute the slowdown primarily to increased demand for certain mature nodes (e.g., 16nm and 65nm for CoWoS processing), which drove 16% and 28% YoY wafer shipment growth in the past two quarters. Meanwhile, the absence of price hikes at these nodes has further weighed on blended ASP growth.

However, we expect robust N3 demand to drive its revenue mix back above 30% from 2Q26 onward (vs. 25% in 1Q26), with N2 contributions also building through the year. We believe the improved mix—both by node and by application (higher HPC weighting)—combined with continued hot/super-hot run demand (carrying 50–100% ASP premiums), should lift blended ASP growth back above 20% YoY in 2026 and 2027, up from 16% in 2025.

Longer term, we expect TSMC to sustain 15–20% annual blended ASP growth, underpinned by its ability to maintain elevated pricing at leading-edge nodes over extended periods, while wafer shipments grow in the 5–10% range in steady state. We see TSMC's revenue growth remaining primarily ASP-driven, with sustained strong pricing helping to deliver improved returns on rising capex and driving better ROI through the AI-driven semiconductor upcycle. This dynamic should also help smooth revenue volatility, as wafer shipments tend to swing more sharply during downturns.

Figure 10: Breakdown of TSMC wafer sales by blended ASP and wafer shipments



Source: J.P. Morgan estimates, Company data.

1Q26 Earnings Review

Table 1: 1Q26 Earnings Review

NT$bn, EPS in NT$, %

	1Q26			Variance		Growth	
NT$ billion	Actual	JPMe	BBG 28 days	vs. JPMe	vs. Consensus	Q/Q	Y/Y
Revenues	**1,134**	**1,138**	**1,128**	**-0%**	**1%**	**8%**	**35%**
Gross profit	751	760	733	-1%	2%	15%	52%
GM (%)	*66.2%*	*66.8%*	*65.0%*	*-52 bps*	*125 bps*	*392 bps*	*746 bps*
Operating profit	659	656	631	0%	4%	17%	62%
OPM (%)	*58.1%*	*57.7%*	*56.0%*	*44 bps*	*213 bps*	*410 bps*	*960 bps*
Pretax profit	688	673	656	2%	5%	16%	60%
Net profit	572	556	550	3%	4%	13%	58%
EPS (NT$)	**22.08**	**21.42**	**21.19**	**3%**	**4%**	**13%**	**58%**

Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates.

Forecast Changes

Table 2: TSMC: Quarterly estimate revisions

	Revised			Prior			Change (%)		
NT$ billion	2Q26E	3Q26E	4Q26E	2Q26E	3Q26E	4Q26E	2Q26E	3Q26E	4Q26E
Revenues (US$ in mn)	**39,813**	**43,999**	**48,246**	**38,845**	**43,527**	**47,067**	**2%**	**1%**	**3%**
Revenues (NT$ in bn)	**1,262**	**1,395**	**1,529**	**1,239**	**1,389**	**1,501**	**2%**	**0%**	**2%**
Gross profit	845	906	999	823	865	943	3%	5%	6%
GM (%)	*67.0%*	*65.0%*	*65.3%*	*66.4%*	*62.3%*	*62.8%*	*51bps*	*270bps*	*254bps*
Operating profit	728	779	863	708	738	809	3%	6%	7%
OPM (%)	*57.7%*	*55.9%*	*56.4%*	*57.1%*	*53.2%*	*53.9%*	*51bps*	*270bps*	*254bps*
Net profit	598	667	738	603	629	689	-1%	6%	7%
EPS (NT$)	**23.07**	**25.71**	**28.46**	**23.27**	**24.25**	**26.57**	**-1%**	**6%**	**7%**
Operating Assumptions									
Capacity ('000 wafers, 12" equ)	4,114	4,113	4,175	4,074	4,134	4,214	1%	-1%	-1%
Shipment ('000 wafers, 12" equ)	4,097	4,166	4,240	3,958	4,133	4,178	4%	1%	1%
Shipment utilization (%)	100%	101%	102%	97%	100%	99%	243bps	129bps	242bps
Wafer ASP (US$)	8,222	8,767	9,349	8,276	8,792	9,339	-1%	0%	0%

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Source: J.P. Morgan estimates.

Table 3: TSMC: Annual estimate revisions

	Revised			Prior			Change (%)		
NT$ billion	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
Revenues (US$ in mn)	**167,960**	**221,550**	**258,379**	**165,453**	**214,293**	**252,989**	**2%**	**3%**	**2%**
Revenues (NT$ in bn)	**5,320**	**7,023**	**8,191**	**5,267**	**6,836**	**8,070**	**1%**	**3%**	**1%**
Gross profit	3,502	4,526	5,330	3,390	4,334	5,177	3%	4%	3%
GM (%)	*65.8%*	*64.4%*	*65.1%*	*64.4%*	*63.4%*	*64.2%*	*144bps*	*105bps*	*92bps*
Operating profit	3,029	3,859	4,554	2,912	3,684	4,414	4%	5%	3%
OPM (%)	*56.9%*	*54.9%*	*55.6%*	*55.3%*	*53.9%*	*54.7%*	*165bps*	*105bps*	*92bps*
Net profit	2,575	3,244	3,818	2,477	3,123	3,732	4%	4%	2%
EPS (NT$)	**99.31**	**125.11**	**147.21**	**95.50**	**120.44**	**143.92**	**4%**	**4%**	**2%**
Operating Assumptions									
Capacity ('000 wafers, 12" equ)	16,446	17,414	17,721	16,446	17,526	18,061	0%	-1%	-2%
Shipment ('000 wafers, 12" equ)	16,676	17,491	17,436	16,151	17,432	17,935	3%	0%	-3%
Shipment utilization (%)	101%	100%	98%	98%	99%	99%	319bps	97bps	-91bps
Wafer ASP (US$)	8,445	10,448	11,929	8,618	10,285	11,681	-2%	2%	2%

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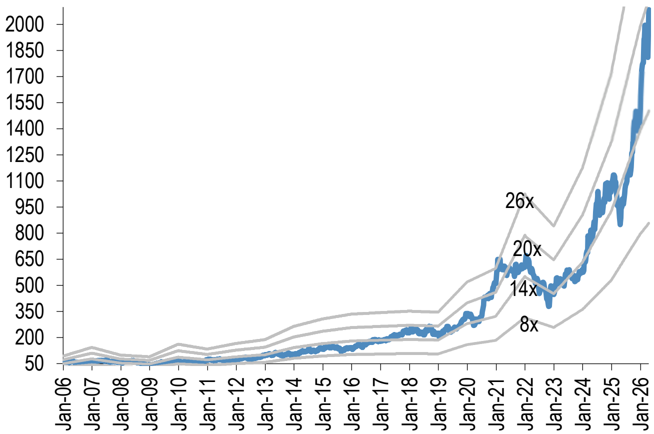
Source: J.P. Morgan estimates.

PE/PB charts

Figure 11: TSMC : 12month forward P/E

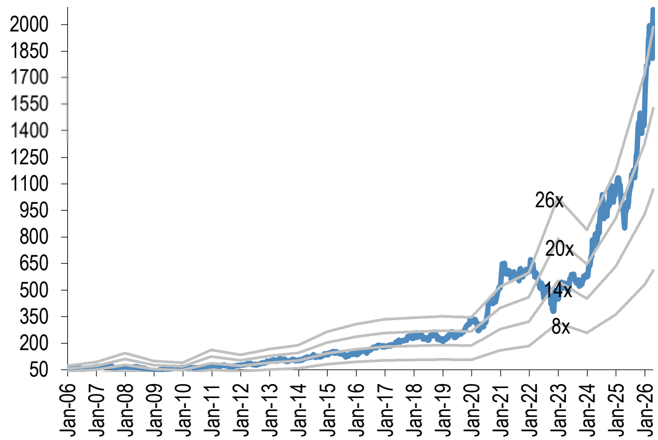
NT$, x



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 12: TSMC : 12month trailing P/E

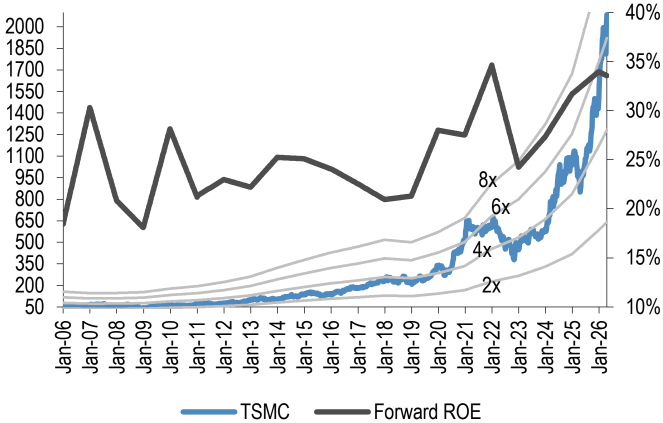
NT$, x



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 13: TSMC : 12month forward P/B and ROE

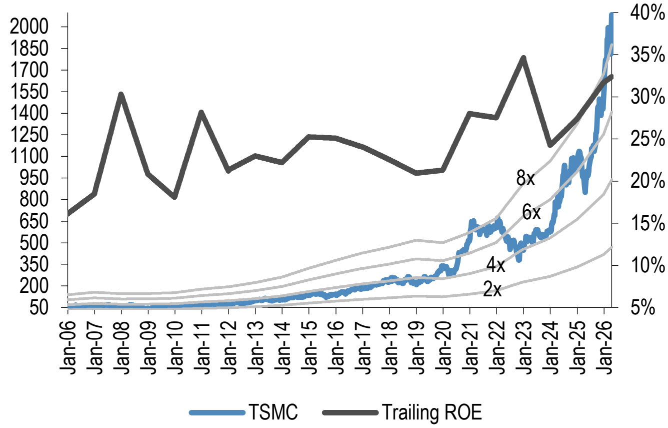
NT$, x, %



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 14: TSMC : 12month trailing P/B and ROE

NT$, x, %



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

1Q26 Earnings Call Highlights

1Q26 Earnings Call Highlights

1Q26 recap

- Revenue $35.9bn, up 6.4% QoQ in USD terms, driven by strong demand of leading edge tech
- GM 66.2%, +3.9ppt QoQ, driven by cost improvement, higher capacity UTR, more favorable FX – exceeding high end guidance, due to higher UTR and better cost improvement
- OPM 58.1%, +4.1ppt QoQ, due to operating leverage
- EPS: NT$22.08
- ROE: 40.5%
- Shipment 4,174k pcs, +5.4% QoQ
- FX 31.59
- Capex: NT$351bn (US$11.1bn)

Revenue mix by process

- N3: 25% of sales

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- N5: 36% of sales
- N7: 13% of sales
- N7 and below – 74% of wafer sales

Revenue mix by platform

- HPC up 20% QoQ (61% of sales)
- Smartphone down 11% QoQ (26% of sales)

2Q26 outlook

- Revenue US$39.0-40.2bn, up 10% QoQ or 32% YoY at midpoint – supported by strong demand in leading edge, mindful of rising component price in Consumer Electronics segment
  - Middle East macro uncertainty – being prudent on planning and focusing on fundamentals

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  - AI related demand robust, gen AI moving to agentic AI, another step-up for token generation
  - Customers and their customers (CSP) giving very strong signals and positive outlook
- GM 65.5-67.5%, driven by higher UTR and continued cost improvement, partially offset by overseas fabs
- OPM: 56.5-58.5%

- Tax 20% in 2Q26 due to undistributed retained earnings, still maintain 17-18% goal for FY26
- FX 31.7

2026 outlook

- FY26 rev – confident on >30% YoY in USD terms
- Initial ramp for 2nm starts to dilute GM in 2H26, between 2-3% for FY26

  - GM dilution in next several years, 2-3% in the early stage and 3-4% in the later stage
  - Costs of certain chemicals price increase due to Middle East conflict – will have some impact on profitability but too early to quantify the impact

- N3 to cross over corporate average GM in 2H26
- FX should be another factor but not controllable
- Material supply is developing multi-source and diversified-based, Helium and H2 sourced from multiple regions and have safety inventory, not expect any near term impact on operations
- Energy – working with Taipower and gov for stable supply, TW gov announced sufficient LNG to at least May, actively securing further supply with diversifying to other regions – not expecting impact in the near term

2026 capex

- High end of $52-56bn – driven by 5G AI and HPC

TW Fab update

- Prioritize land in TW to support ramp on new node
- New N2 node – already enter MP in 4Q25 with good yield, ramping with successfully in multiple phases at Hsinchu and Kaohsiung for smartphone and HPC
- Expect N2 family to be large and long lasting node for TSMC

N3

- AI demand – stepping up capex to increase N3 capacity, global capacity plan to support the pipeline for Smartphone HPC AI (incl HBM base die and auto, IoT)
- Adding new N3 in Tainan Gigafab, planning 1H27 MP
- Arizona – 2^{nd} fab also using N3, complete construction and MP in 2H27
- Japan – plan N3 in 2^{nd} fab and MP in 2028
- Continue to convert N5 to N3 in TW, driving better productivity in all locations
- N7 and below – using multiple levers to max support to all customers at all platforms
- We don’t pick and choose customers

Mature node

- Unchanged plan, focus on building high yield for specialized capacity
- Increasing capacity in JASM fab 1 for CMOS image sensors

- ESMC fabs for auto and industrials
- Winding down on fab 2 (6”) and fab 5 (8”) for GAN – using the available space for leading edge applications
- Without Fab 2 and Fab 5 – still have enough to fully support our existing customers

A14

- Featuring a second-generation nanosheet transistor structure
- Deliver four node stride from N2, power benefit to address in high performance and efficient power demand, vs. N2, 10-15% higher improvement in speed at the same power, 25-30% power improvement at same speed, close to 20% chip density gains – schedule on track, 2028 MP

Q&A

N3 GM – application driver for future demand, GM outlook to what level?

- Applications still HPC AI
- GM – expecting GM to reach company average in 2H26, no numbers to share, but after fully depreciated, GM should be generally high

Capex – what is the incremental confidence source of the step-up?

- Demand is robust from HPC and AI, trying to speed up and pull in all the equipment, but still tight supply as demand continues to increase – that’s why we're stepping up the forecast

Supply demand– expectation on how long the supply demand constraint to last, plan to build more clean room space?

- Demand continues to be robust and checks with customers and their customers (CSP) giving very positive outlook, we need to speed up our equipment and pulling forward vs. the forecasted schedule
- When to meet these demand? Timeframe? – 2-3 years to build new fab, but takes time to ramp up, should continue to be really tight - that’s why we are building new N3 fabs to meet the demand

Competition – view on the initiatives from competitors (Terafab), Samsung trying to build chips on their own?

- Both Intel and TSLA are TSMC customers – also our competitors
- Intel as a formidable competitor, do not underestimate them
- No shortcuts, they need tech leadership, manufacturing excellence, customer trust, and services
    - 2-3 years to build new fabs and 1-2 years to ramp up – this is fundamental
- Try to win back – still our customer and very confident in our tech
- Tight capacity, is it why we lose customers? – takes 2-3 years to build new fabs, we are also building to meet strong demand, no shortcuts

Competition – what is TSMC's strategy on the competition from EMIB or open up their compute die to Intel to do the packaging?

- TSMC supplying largest reticle size packaging, understand our competitors offer competitive packaging, customers have more choices, we don't leave biz on the table
- Developing very large reticle size tech and working with customers - so far so good
  - What is the tech you refer to? – have large reticle size CoWoS and working on CoPoS, making sure we have enough capacity with reasonable cost, already having CoPoS pilot line, expecting MP in the next few years later
  - Now the main approach is still the large size CoWoS with system on wafer tech – the best option in the market

Capex – long term guidance in next 3 years

- No numbers to share – past 3 years, total capex $101bn, 2026 toward the high end of $56bn, which is >50% for the past 3yrs aggregate, showing strong demand from AI mega trend, capex next 3 yrs will be significantly higher than last 3 years
- Any bottlenecks with long leading time? - always working with suppliers as partners, ASML, Applied Materials, Lam Research – happy with their support

Capex – from 2024-2026, keeping capital intensity at 30+%; in coming few years, given strong ramp, what is the revenue growth vs. capex growth?

- Past 3 years, revenue growth outpaced capex growth, should continue to see this trend if we do the things right
- Not expecting a sudden surge in the capital intensity
- Expand more on capex due to competition for not enough capacity? – preparation the capacity to meet the demand, not due to other competitors, it's due to customer demand, so we plan our capex

FY26 revenue guidance – guiding 30+% growth, how much more upside, any impact from Smartphone?

- Memory price hikes have some impact on the end market for PC and Smartphone, seeing softer market, but high end Smartphone doing better, this is favorable to TSMC advantage
- How much higher vs. 30% YoY? Will share that in Jul

Margin – any change in the margin structure given strong demand for long-term?

- Have revised up long term GM and ROE from 2024-2029, GM will be 56% and higher through the cycle, looking ROE at high 20% - already higher than before
- Long-term planning is ongoing, will update when there are changes

US fab – Mid-term and long-term opportunities and becoming more strategic and what is the economics and margin outlook?

- Acquired 2nd land, because we need it, to meet multi-year demand for the leading edge customers
- Working hard to speed up and already gaining a lot of experience in Arizona, more confident than 2025 to make good progress and moving aggressively forward,

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expecting to improve the cost structure

Profitability – is it fully reflect the TSMC value? What is the benchmark to look at for GM and OPM?

- Pricing strategy – view customers as our partners, we know our value and position, view partners as important biz partners, don't change pricing dramatically, make sure customers are successful and earn our value – expand our capacity to support them

- Fundamentals – our customers need to be successful, we grow together
- Customers are our partners

AI growth – mid to high 50s CAGR, given strong growth in tokens, seeing any changes for AI growth in the coming years?

- Very strong demand – continue to see positive signals from customers
- Changes to AI growth? – seeing strong demand, towards higher 50s

Advanced Packaging – business model with OSAT, saw various solutions from OSAT, how do we work with customers to align with advanced packaging demand?

- Priority is to support customers – making sure their product can be met by TSMC front end and advanced packaging
- Advanced packaging capacity is very tight as well – need to work with OSATs and increase capacity for our customers, we are also working our capacity as well

Advanced Packaging – AI chips are growing with very large die size, potential tech challenges are from warpage, following roadmap for SoIC and CoPoS can solve these issues?

- TSMC has a lot of experience, as we supply all the leading edge packaging, continue to increase die size to meet the challenge from mechanical stress like warpage and thermal
- The harder the better – it's our strength, we can work with the customers to solve them
- SoIC, what is the development? – working with our customers and meet their demands

AI rev – definition of AI accelerators? Will we include CPUs, any historical numbers to share?

- CPUs becoming more important – but TSMC not able to identify which CPUs go to where

- Still not including CPU in AI HPC calculations – someday later might consider

Competition – LPU at Samsung foundry? How does TSMC win back LPU biz? Any plan?

- Very confident in our tech and working hard to capture every piece of the biz

Figure 15: TSMC: Earnings model

	FY25				FY26E				FY27E				FY28E							
NT$ in billions, year-end December	1Q	2Q	3Q	4Q	1Q	2QE	3QE	4QE	1QE	2QE	3QE	4QE	1QE	2QE	3QE	4QE	FY25	FY26	FY27E	FY28E
Sales (NT$ in billions)	**839.3**	**933.8**	**989.9**	**1,046.1**	**1,134.1**	**1,262.1**	**1,394.8**	**1,529.4**	**1,550.1**	**1,700.6**	**1,856.2**	**1,916.2**	**1,884.4**	**2,027.4**	**2,167.6**	**2,111.2**	**3,809.1**	**5,320.4**	**7,023.1**	**8,190.6**
Sales (US$ in millions)	**25,525**	**30,074**	**33,097**	**33,731**	**35,898**	**39,813**	**43,999**	**48,246**	**48,899**	**53,648**	**58,556**	**60,447**	**59,446**	**63,956**	**68,380**	**66,598**	**122,456**	**167,960**	**221,550**	**258,379**
COGS	-345.9	-386.4	-401.4	-394.1	-382.8	-417.0	-488.6	-530.2	-562.2	-604.9	-652.0	-677.9	-667.5	-706.5	-744.2	-742.9	-1,527.8	-1,818.6	-2,497.1	-2,861.1
Depreciation	-175.1	-188.1	-162.8	-162.1	-165.5	-188.7	-220.9	-230.1	-236.7	-253.6	-268.0	-278.4	-285.6	-301.0	-314.0	-326.0	-688.1	-805.2	-1,036.7	-1,226.5
Gross profit	**493.4**	**547.4**	**588.5**	**652.0**	**751.3**	**845.1**	**906.2**	**999.2**	**987.9**	**1,095.7**	**1,204.2**	**1,238.2**	**1,216.9**	**1,320.9**	**1,423.4**	**1,368.2**	**2,281.3**	**3,501.8**	**4,526.1**	**5,329.6**
Operating expense	-86.3	-83.9	-87.9	-87.1	-92.3	-117.4	-126.9	-136.1	-147.3	-161.6	-176.3	-182.0	-177.1	-190.6	-202.7	-204.8	-345.2	-472.7	-667.2	-775.2
SG&A	-29.8	-22.7	-24.1	-22.2	-24.6	-30.3	-33.5	-36.7	-38.8	-42.5	-46.4	-47.9	-45.2	-48.7	-53.1	-50.7	-98.8	-125.0	-175.6	-197.7
R&D	-56.5	-61.3	-63.7	-64.9	-67.8	-87.1	-93.5	-99.4	-108.5	-119.0	-129.9	-134.1	-131.9	-141.9	-149.6	-154.1	-246.4	-347.7	-491.6	-577.5
Operating (EBIT)	**407.1**	**463.4**	**500.7**	**564.9**	**659.0**	**727.7**	**779.3**	**863.1**	**840.6**	**934.2**	**1,027.9**	**1,056.2**	**1,039.8**	**1,130.4**	**1,220.8**	**1,163.4**	**1,936.1**	**3,029.1**	**3,858.9**	**4,554.4**
Net interest income (expense)	22.2	21.5	23.2	26.5	26.1	19.5	23.3	25.4	19.0	19.7	20.6	22.6	20.7	21.9	19.6	21.6	93.4	94.3	81.9	83.9
Net other income (expense)	1.6	8.1	1.5	1.0	2.7	1.0	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	12.2	5.8	4.3	4.4
Pre-tax profit	430.9	493.0	525.4	592.4	687.8	748.2	803.6	889.5	860.7	954.9	1,049.5	1,079.9	1,061.6	1,153.4	1,241.5	1,186.2	2,041.7	3,129.2	3,945.1	4,642.7
Tax credit (expense)	-70.2	-95.5	-73.6	-86.9	-115.0	-149.6	-136.6	-151.2	-146.3	-191.0	-178.4	-183.6	-180.5	-230.7	-211.1	-201.7	-326.3	-552.5	-699.3	-823.9
Net profit	**361.6**	**398.3**	**452.3**	**505.7**	**572.5**	**598.2**	**666.7**	**738.0**	**714.1**	**763.6**	**870.8**	**896.0**	**880.8**	**922.4**	**1,030.1**	**984.2**	**1,717.9**	**2,575.4**	**3,244.5**	**3,817.5**
Adjusted O/S (B)	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9
FD EPS (NT$)	**13.94**	**15.36**	**17.44**	**19.50**	**22.08**	**23.07**	**25.71**	**28.46**	**27.54**	**29.45**	**33.58**	**34.55**	**33.97**	**35.57**	**39.72**	**37.95**	**66.24**	**99.31**	**125.11**	**147.21**
Key assumption																				
Capacity ('000 wafers, 12" equ)	3,819	3,874	3,894	3,974	4,044	4,114	4,113	4,175	4,222	4,307	4,407	4,477	4,504	4,459	4,409	4,349	15,561	16,446	17,414	17,721
Shipment ('000 wafers, 12" equ)	3,259	3,718	4,085	3,961	4,174	4,097	4,166	4,240	4,160	4,327	4,533	4,471	4,357	4,425	4,470	4,184	15,023	16,676	17,491	17,436
Shipment utilization (%)	85	96	105	100	103	100	101	102	99	100	103	100	97	99	101	96	97	101	100	98
Wafer ASP (US$)	6,663	6,891	7,046	7,344	7,422	8,222	8,767	9,349	9,790	10,322	10,697	10,929	11,118	11,718	12,321	12,578	7,003	8,445	10,448	11,929
Margins (%)																				
Gross	58.8	58.6	59.5	62.3	66.2	67.0	65.0	65.3	63.7	64.4	64.9	64.6	64.6	65.2	65.7	64.8	59.9	65.8	64.4	65.1
Operating	48.5	49.6	50.6	54.0	58.1	57.7	55.9	56.4	54.2	54.9	55.4	55.1	55.2	55.8	56.3	55.1	50.8	56.9	54.9	55.6
EBITDA	69.4	69.8	67.0	69.5	72.7	72.6	71.7	71.5	69.5	69.8	69.8	69.7	70.3	70.6	70.8	70.5	68.9	72.1	69.7	70.6
Net	43.1	42.7	45.7	48.3	50.5	47.4	47.8	48.3	46.1	44.9	46.9	46.8	46.7	45.5	47.5	46.6	45.1	48.4	46.2	46.6
Growth (%, Q/Q)																				
Shipment	-5	14	10	-3	5	-2	2	2	-2	4	5	-1	-3	2	1	-6				
ASP	-2	3	2	4	1	11	7	7	5	5	4	2	2	5	5	2				
Sales (in NT$ terms)	-3	11	6	6	8	11	11	10	1	10	9	3	-2	8	7	-3				
Sales (in US$ terms)	-5	18	10	2	6	11	11	10	1	10	9	3	-2	8	7	-3				
Gross profit	-4	11	8	11	15	12	7	10	-1	11	10	3	-2	9	8	-4				
EBIT	-4	14	8	13	17	10	7	11	-3	11	10	3	-2	9	8	-5				
Net profit	-4	10	14	12	13	4	11	11	-3	7	14	3	-2	5	12	-4				
EPS	-4	10	14	12	13	4	11	11	-3	7	14	3	-2	5	12	-4				
Growth (%, Y/Y)																				
Shipment	8	19	22	16	28	10	2	7	0	6	9	5	5	2	-1	-6	16	11	5	0
ASP	22	17	18	8	11	19	24	27	32	26	22	17	14	14	15	15	16	21	24	14
Sales (in NT$ terms)	42	39	30	20	35	35	41	46	37	35	33	25	22	19	17	10	32	40	32	17
Sales (in US$ terms)	35	44	41	26	41	32	33	43	36	35	33	25	22	19	17	10	36	37	32	17
Gross profit	57	53	34	27	52	54	54	53	31	30	33	24	23	21	18	10	40	54	29	18
EBIT	63	62	39	33	62	57	56	53	28	28	32	22	24	21	19	10	46	56	27	18
Net profit	60	61	39	35	58	50	47	46	25	28	31	21	23	21	18	10	46	50	26	18
EPS	60	61	39	35	58	50	47	46	25	28	31	21	23	21	18	10	46	50	26	18

Investment Thesis, Valuation and Risks

TSMC *(Overweight; Price Target: NT$2500.0)*

Investment Thesis

We expect TSMC's structural growth drivers to remain very strong as leading-edge supply (N4, N3, and N2) stays tight well into 2027 or early 2028, given accelerating AI demand, continued growth in N3, N2 and advanced back-end revenues, and long lead time for capacity build despite accelerated capex. We believe the set-up for TSMC stock looks quite favorable in the next few quarters, with potential upside to its long-term capex targets, 2026 revenue guidance, and GMs beating due to high UTR and ASP uptick. Given industry-wide shortage, we expect competitive noise to persist, but achieving technology and manufacturing milestones and winning customer acceptance would be the key challenge for the chasing pack, given TSMC tech and execution leadership and 3-5 year window for design and manufacturing ramp.

Valuation

Our Dec-26 PT of NT$2,500 is based on ~20x 12-month forward P/E and reflects better GMs and slightly faster growth. Our target multiple is higher than TSMC's five-year average historical multiple.

Risks to Rating and Price Target

Key downside risks to our rating and price target include: (1) the debate on the duration of the AI capex growth cycle and (2) any negative impact from weak PC/Smartphone demand in 2H26.

TSMC: Summary of Financials

Income Statement - Annual	FY24A	FY25A	FY26E	FY27E
Revenue	**2,894,308**	**3,809,054**	**5,320,358**	**7,023,133**
COGS	(1,269,954)	(1,527,760)	(1,818,552)	(2,497,076)
Gross profit	**1,624,354**	**2,281,294**	**3,501,806**	**4,526,057**
SG&A	(98,119)	(98,775)	(125,042)	(175,578)
Adj. EBITDA	**1,984,850**	**2,624,188**	**3,834,219**	**4,895,538**
D&A	(662,797)	(688,096)	(805,157)	(1,036,679)
Adj. EBIT	**1,322,053**	**1,936,092**	**3,029,063**	**3,858,859**
Net Interest	76,718	93,369	94,287	81,911
Adj. PBT	**1,405,839**	**2,041,663**	**3,129,180**	**3,945,053**
Tax	*(233,407)*	*(326,266)*	*(552,480)*	*(699,306)*
Minority Interest	836	2,486	(1,284)	(1,284)
Adj. Net Income	**1,173,268**	**1,717,883**	**2,575,416**	**3,244,463**
Reported EPS	45.24	66.24	99.31	125.11
Adj. EPS	**45.24**	**66.24**	**99.31**	**125.11**
DPS	**16.50**	**22.00**	**25.50**	**34.00**
Payout ratio	51.0%	48.6%	38.5%	34.2%
Shares outstanding	25,933	25,933	25,933	25,933

Balance Sheet & Cash Flow Statement	FY24A	FY25A	FY26E	FY27E
Cash and cash equivalents	2,422,020	3,068,595	4,466,940	5,603,522
Accounts receivable	272,088	282,059	419,013	524,978
Inventories	287,869	288,109	419,013	524,978
Other current assets	106,376	178,367	135,947	170,326
Current assets	**3,088,352**	**3,817,131**	**5,440,913**	**6,823,804**
PP&E	3,234,980	3,691,841	4,356,199	5,380,021
LT investments	149,040	172,370	169,710	173,992
Other non current assets	219,565	251,682	274,192	274,192
Total assets	**6,691,938**	**7,933,024**	**10,241,015**	**12,652,009**
Short term borrowings	59,858	136,926	156,242	156,242
Payables	74,227	84,330	108,943	121,795
Other short term liabilities	1,130,441	1,236,763	1,359,466	1,419,384
Current liabilities	**1,264,525**	**1,458,019**	**1,624,651**	**1,697,421**
Long-term debt	958,429	896,062	860,026	860,026
Other long term liabilities	145,408	118,147	132,277	107,740
Total liabilities	**2,368,362**	**2,472,229**	**2,616,954**	**2,665,187**
Shareholders' equity	4,288,545	5,419,596	7,582,631	9,945,393
Minority interests	35,031	41,199	41,429	41,429
Total liabilities & equity	**6,691,938**	**7,933,024**	**10,241,015**	**12,652,009**
BVPS	**165.4**	**209.0**	**292.4**	**383.5**
y/y Growth	*24.0%*	*26.4%*	*39.9%*	*31.2%*
Net debt/(cash)	(1,403,733)	(2,035,607)	(3,450,672)	(4,587,254)
Cash flow from operating activities	**1,975,661**	**2,437,716**	**3,303,735**	**4,108,887**
o/w Depreciation & amortization	662,797	688,096	805,157	1,036,679
o/w Changes in working capital	(90,016)	(108)	(243,245)	(199,077)
Cash flow from investing activities	**(928,044)**	**(1,200,404)**	**(1,489,365)**	**(2,064,783)**
o/w Capital expenditure	(958,198)	(1,272,219)	(1,773,870)	(2,060,500)
as % of sales	*33.1%*	*33.4%*	*33.3%*	*29.3%*
Cash flow from financing activities	**(313,242)**	**(590,737)**	**(416,026)**	**(907,522)**
o/w Dividends paid	(428,731)	(573,003)	(659,992)	(880,418)
o/w Net debt issued/(repaid)	90,711	14,701	(16,720)	0
Net change in cash	**734,375**	**646,575**	**1,398,345**	**1,136,582**
Adj. Free cash flow to firm	**1,142,360**	**1,292,759**	**1,834,220**	**2,048,387**
y/y Growth	*164.7%*	*13.2%*	*41.9%*	*11.7%*

Income Statement - Quarterly	1Q26A	2Q26E	3Q26E	4Q26E
Revenue	**1,134,103**	**1,262,076**	**1,394,779**	**1,529,399**
COGS	(382,808)	(417,008)	(488,563)	(530,173)
Gross profit	**751,295**	**845,069**	**906,217**	**999,226**
SG&A	(24,572)	(30,290)	(33,475)	(36,706)
Adj. EBITDA	**824,416**	**916,413**	**1,000,144**	**1,093,246**
D&A	(165,450)	(188,718)	(220,853)	(230,136)
Adj. EBIT	**658,966**	**727,695**	**779,292**	**863,109**
Net Interest	26,146	19,464	23,304	25,373
Adj. PBT	**687,800**	**748,201**	**803,643**	**889,536**
Tax	(114,999)	(149,640)	(136,619)	(151,221)
Minority Interest	(321)	(321)	(321)	(321)
Adj. Net Income	**572,480**	**598,240**	**666,703**	**737,994**
Reported EPS	22.08	23.07	25.71	28.46
Adj. EPS	**22.08**	**23.07**	**25.71**	**28.46**
Shares outstanding	25,932	25,932	25,932	25,932

Ratio Analysis	FY24A	FY25A	FY26E	FY27E
Gross margin	56.1%	59.9%	65.8%	64.4%
EBITDA margin	68.6%	68.9%	72.1%	69.7%
EBIT margin	45.7%	50.8%	56.9%	54.9%
Net profit margin	40.5%	45.1%	48.4%	46.2%
ROE	30.3%	35.4%	39.6%	37.0%
ROA	19.2%	23.5%	28.3%	28.3%
ROCE	22.7%	27.7%	33.1%	32.5%
SG&A/Sales	3.4%	2.6%	2.4%	2.5%
Net debt/equity	NM	NM	NM	NM
P/E (x)	46.1	31.5	21.0	16.7
P/BV (x)	12.6	10.0	7.1	5.4
EV/EBITDA (x)	26.5	19.8	13.2	10.1
Dividend Yield	0.8%	1.1%	1.2%	1.6%
Sales/Assets (x)	47.4%	52.1%	58.5%	61.4%
Interest cover (x)	NM	NM	NM	NM
Operating leverage	128.3%	147.0%	142.3%	85.6%
Revenue y/y Growth	33.9%	31.6%	39.7%	32.0%
EBITDA y/y Growth	36.5%	32.2%	46.1%	27.7%
Tax rate	16.6%	16.0%	17.7%	17.7%
Adj. Net Income y/y Growth	39.9%	46.4%	49.9%	26.0%
EPS y/y Growth	39.9%	46.4%	49.9%	26.0%
DPS y/y Growth	26.9%	33.3%	15.9%	33.3%

Source: Company reports and J.P. Morgan estimates.
Note: NT$ in millions (except per-share data).Fiscal year ends Dec. o/w - out of which

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